Finite Elements for Linear and Non Linear Mechanics

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Chapter 1

Introduction

This document aims to provide a theoretical explanation for the finite element method (FEM) for solid mechanics problems. The document begins by presenting all the necessary theoretical background, including the definition of many kinematics and stress related variables. From those, the available material constitutive models are presented, which are used to relate the kinematics and stress variables. Then, equation to be solved is derived by using the principle of virtual work. It is then discretized by using the finite element method. Finally, both solution methods are explained. Addionally, the document will present an overview of the code structure and all of its components.

This project is based on a academic projet I made in 2025 at Centrale Nantes. Based on that, the approach has been adapted to be used in a more general way, and the code has been rewritten for better comprehension, modularity and optimize performance - obtained mainly through the use of numpy array instead of loops. Addionally, the book *"Nonlinear Continuum Mechanics for Finite Element Analysis"* by Javier Bonet and Richard D. Wood was used for theoretical reference.

Chapter 2

Kinematics and Stresses

In this section, the kinematics and the different stress tensors used in this paper are defined. The relation between them is also given. Finally, the definition of the strain energy density is given as well as its relation with the different stress tensors. These definitions are used throughout the paper and are important for the understanding of the following sections.

2.1 Kinematics

The kinematics of a body is defined by the transformation between the initial configuration and the current configuration. This transformation can be defined as stated by Equation 2.1. Thoughout this paper, the initial configuration will be denoted by capital letters while the current configuration will be denoted by lowercase letters.

$$\vec{x} = \vec{\phi}(\vec{X},\, t) \tag{2.1}$$

Based on this transformation, it is possible to define the transformation gradient (Equation 2.2) and the Jacobian of the transformation (Equation 2.3).

$$\bar{\bar{\mathrm{F}}} = \bar{\bar{\nabla}}_0 \vec{x} = \frac{\partial \vec{x}}{\partial \vec{X}} \tag{2.2}$$

$$\mathrm{J} = \det \bar{\bar{\mathrm{F}}} \tag{2.3}$$

Based on the tensor $\bar{\bar{\mathrm{F}}}$, it is possible to define multiple tensors to represent the defromation. In this codes, two of then are used: the right Cauchy-Green deformation tensor $\bar{\bar{\mathrm{C}}}$ and the Green-Lagrange strain tensor $\bar{\bar{\mathrm{E}}}$ as given by Equations 2.4 and 2.5 respectively.

$$\bar{\bar{\mathrm{C}}} = \bar{\bar{\mathrm{F}}}^T \bar{\bar{\mathrm{F}}} \tag{2.4}$$

$$\bar{\bar{\mathrm{E}}} = \frac{1}{2}\left(\bar{\bar{\mathrm{C}}} - \bar{\bar{\mathrm{I}}}\right) = \frac{1}{2}\left(\bar{\bar{\mathrm{F}}}^T \bar{\bar{\mathrm{F}}} - \bar{\bar{\mathrm{I}}}\right) \tag{2.5}$$

2.2 Stresses

Four different stress tensors are used throughout this paper and are listed below:

- $\bar{\bar{\sigma}}$: Cauchy stress tensor
- $\bar{\bar{\mathcal{T}}}$: kirchhoff stress tensor
- $\bar{\bar{\mathrm{P}}}$: First Piola-Kirchhoff stress tensor
- $\bar{\bar{\mathrm{S}}}$: Second Piola-kirchhoff stress tensor

They are all stress' representations but under different configurations. They can all be related under the following equation:

$$\bar{\bar{\sigma}} = \mathrm{J}^{-1} \cdot \bar{\bar{\mathcal{T}}} = \mathrm{J}^{-1} \cdot \bar{\bar{\mathrm{P}}}\bar{\bar{\mathrm{F}}}^T = \mathrm{J}^{-1} \cdot \bar{\bar{\mathrm{F}}}\bar{\bar{\mathrm{S}}}\bar{\bar{\mathrm{F}}}^T \tag{2.6}$$

2.3 Strain Energy Density

The strain energy density is defined as:

$$\Psi(\bar{\bar{\mathrm{F}}}, \vec{X}) = \int \bar{\bar{\mathrm{P}}} : \dot{\bar{\bar{\mathrm{F}}}}\,\mathrm{dt} \tag{2.7}$$

From this definition, a new definition for $\bar{\bar{\mathrm{P}}}$ can be stated:

$$\bar{\bar{\mathrm{P}}} = \frac{\partial \Psi}{\partial \bar{\bar{\mathrm{F}}}} \tag{2.8}$$

This can also be written as a function of the tensor $\bar{\bar{\mathrm{C}}}$ as given below:

$$\Psi(\bar{\bar{\mathrm{F}}}, \vec{X}) = \Psi(\bar{\bar{\mathrm{C}}}, \vec{X}) = \int \frac{1}{2}\bar{\bar{\mathrm{S}}} : \dot{\bar{\bar{\mathrm{C}}}}\,\mathrm{dt} \tag{2.9}$$

This means the $\bar{\bar{\mathrm{S}}}$ can be written as:

$$\bar{\bar{\mathrm{S}}} = 2\frac{\partial \Psi}{\partial \bar{\bar{\mathrm{C}}}} = \frac{\partial \Psi}{\partial \bar{\bar{\mathrm{E}}}} \tag{2.10}$$

Chapter 3

Material Behavior

3.1 Linear Materials

3.1.1 Linear Elastic Isotropic

This model uses the hypothesis of small displacements under a elastic regime. This is mainly done by assuming that:

$$\bar{\bar{\nabla}}\vec{u} \approx \bar{\bar{\nabla}}_0\vec{u} \qquad \text{and} \qquad \|\bar{\bar{\nabla}}\vec{u}\| \approx 0 \tag{3.1}$$

This leads to the following conclusions:

$$\begin{cases} \bar{\bar{\mathrm{F}}} & = \bar{\bar{\mathrm{I}}} + \bar{\bar{\nabla}}_0\vec{u} \approx \bar{\bar{\mathrm{I}}} \\ \mathrm{J} & = \det \bar{\bar{\mathrm{F}}} \approx 1 \end{cases} \qquad \Longrightarrow \qquad \bar{\bar{\sigma}} \approx \bar{\bar{\mathcal{T}}} \approx \bar{\bar{\mathrm{P}}} \approx \bar{\bar{\mathrm{S}}}$$

Under this hypothesis, the strain ($\bar{\bar{\varepsilon}}$) is used to model the material. It is defined as:

$$\bar{\bar{\varepsilon}} = \frac{1}{2}\left(\bar{\bar{\nabla}}\vec{u} + \bar{\bar{\nabla}}\vec{u}^T\right) \tag{3.2}$$

Using this, both the strain energy density and Cauchy stress can be defined as:

$$\Psi = \frac{1}{2}\lambda\ (tr(\bar{\bar{\varepsilon}}))^2 + \mu\,\bar{\bar{\varepsilon}} : \bar{\bar{\varepsilon}} \tag{3.3}$$

$$\bar{\bar{\sigma}} = \lambda\, tr(\bar{\bar{\varepsilon}})\bar{\bar{\mathrm{I}}} + 2\mu\,\bar{\bar{\varepsilon}} \tag{3.4}$$

The Equation 3.4 is also known as the generalized Hooke's law for isotropic materials. This is written as a function of the stiffness tensor, which is given by:

$$\bar{\bar{\sigma}} = \bar{\bar{\bar{\bar{\mathcal{C}}}}} : \bar{\bar{\varepsilon}} \qquad \text{with} \qquad \mathcal{C}_{ijkl} = \lambda\,\delta_{ij}\delta_{kl} + \mu\,(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \tag{3.5}$$

This model is characterized by two parameters (λ and μ) which are called the Lamé coefficients. They can be written as a function of the Young's modulus (E) and Poisson coefficient (ν) as follows:

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \tag{3.6}$$

$$\mu = \frac{E}{2(1+\nu)} \tag{3.7}$$

3.2 Non Linear Materials

3.2.1 Saint Venant-Kirchhoff

This model can be is often used for moderate deformations under a elastic regime. Unlike the Linear Elastic Isotropic model, the present framework does not uses the small displacement hypothesis and therefore is no longer a linear model.

Similarly, the model uses the strain energy density to correlate the stress and kinematics of the material. Within this context, the strain energy and the second Piola-Kirchhoff stress tensor are give below:

$$\Psi = \frac{1}{2}\lambda \left(tr(\bar{\bar{\mathrm{E}}})\right)^2 + \mu\, \bar{\bar{\mathrm{E}}} : \bar{\bar{\mathrm{E}}} \tag{3.8}$$

$$\bar{\bar{\mathrm{S}}} = \lambda\, tr(\bar{\bar{\mathrm{E}}})\bar{\bar{\mathrm{I}}} + 2\mu\, \bar{\bar{\mathrm{E}}} \tag{3.9}$$

3.2.2 Compressible Neo-Hookean

Unlike the Saint Venant-Kirchhoff model, this model can be employed for large deformations under a elastic regime. It considers a compressible hyper-elastic material that follows the following equations:

$$\Psi = \frac{1}{2}\mu \left(tr(\bar{\bar{\mathrm{C}}}) - 3\right) + \mu\, \ln \mathrm{J} + \frac{1}{2}\lambda\, (\ln \mathrm{J})^2 \tag{3.10}$$

$$\bar{\bar{\mathrm{S}}} = \mu \left(\bar{\bar{\mathrm{I}}} - \bar{\bar{\mathrm{C}}}^{-1}\right) - \lambda\, \ln \mathrm{J}\, \bar{\bar{\mathrm{C}}}^{-1} \tag{3.11}$$

Chapter 4

Finite Element Formulation

4.1 Equilibrium Equation

The equation to be solved is the **equilibrium equation for a stationary regime**, which is given by:

$$\vec{\mathrm{div}}\bar{\bar{\sigma}} + \vec{f} = \vec{0} \tag{4.1}$$

However, the equation above is in its strong formulation. For the finite elements methods, it is far more useful to convert it to its weak form. This allows for the application of boundary conditions directly in the equation. It can be done by using the virtual work theorem:

$$\vec{\mathrm{div}}\bar{\bar{\sigma}} + \vec{f} = \vec{0} \qquad \Longrightarrow \qquad \delta\mathcal{W} = \int_v \left(\vec{\mathrm{div}}\bar{\bar{\sigma}} + \vec{f}\right) \cdot \delta\vec{u}\,\mathrm{dv} = 0 \quad \forall\,\delta\vec{u} \Longrightarrow$$

$$\Longrightarrow \qquad \delta\mathcal{W} = \int_v \vec{\mathrm{div}}\bar{\bar{\sigma}} \cdot \delta\vec{u}\,\mathrm{dv} + \int_v \vec{f} \cdot \delta\vec{u}\,\mathrm{dv} = 0 \quad \forall\,\delta\vec{u}$$

This can be further developed by applying the following rule (Equation 4.2) along side with the divergence theorem (Equation 4.3).

$$\vec{\mathrm{div}}\bar{\bar{A}} \cdot \vec{b} = \vec{\mathrm{div}}\left(\bar{\bar{A}} \cdot \vec{b}\right) - \bar{\bar{A}} : \vec{b} \tag{4.2}$$

$$\int_v \mathrm{div}\left(\vec{a}\right)\,\mathrm{dv} = \int_s \vec{a} \cdot \vec{n}\,\mathrm{ds} \tag{4.3}$$

By applying both relation on the left hand term of the equation, it is possible to obtain:

$$\delta\mathcal{W} = \int_v \bar{\bar{\sigma}} : \bar{\bar{\nabla}}\delta\vec{u}\,\mathrm{dv} - \left(\int_v \vec{f} \cdot \delta\vec{u}\,\mathrm{dv} + \int_s \vec{t} \cdot \delta\vec{u}\,\mathrm{ds}\right) = 0 \quad \forall\,\delta\vec{u} \tag{4.4}$$

Where:

$$\vec{t} = \bar{\bar{\sigma}} \cdot \vec{n}$$

The next step is to rewrite Equation 4.4 with respect to the initial configuration. This can be done by using the Jacobian of the transformation defined by Equation 2.3. Using this, it is possible to rewrite Equation 4.4 as follows:

$$\int_V \bar{\bar{\sigma}} : \bar{\bar{\nabla}} \delta\vec{u}\, J \,\mathrm{dV} - \left(\int_V \vec{f} \cdot \delta\vec{u}\, J \,\mathrm{dV} + \int_S \vec{t} \cdot \delta\vec{u} \left(\frac{\partial s}{\partial S} \right) \mathrm{dS} \right) = 0 \quad \forall\, \delta\vec{u} \qquad \Longrightarrow$$

$$\Longrightarrow \qquad \int_V \underbrace{\bar{\bar{\mathcal{T}}}}_{\mathrm{J}\bar{\bar{\sigma}}} : \bar{\bar{\nabla}} \delta\vec{u} \,\mathrm{dV} - \left(\int_V \underbrace{\vec{f_0}}_{J\vec{f}} \cdot \delta\vec{u} \,\mathrm{dV} + \int_S \underbrace{\vec{t_0}}_{\left(\frac{\partial s}{\partial S}\right)\vec{t}} \cdot \delta\vec{u} \,\mathrm{dS} \right) = 0 \quad \forall\, \delta\vec{u}$$

Furthermore, the gradient can be written as:

$$\bar{\bar{\nabla}} \delta\vec{u} = \frac{\partial \delta\vec{u}}{\partial \vec{x}} = \frac{\partial \delta\vec{u}}{\partial \vec{X}} \cdot \frac{\partial \vec{X}}{\partial \vec{x}} = \bar{\bar{\nabla}}_0 \delta\vec{u} \cdot \bar{\bar{\mathrm{F}}}^{-1}$$

Finally, using that:

$$\bar{\bar{A}} : \left(\bar{\bar{B}} \bar{\bar{C}} \right) = \left(\bar{\bar{A}} \bar{\bar{C}}^T \right) : \bar{\bar{B}}$$

It is possible to obtain the final equation:

$$\delta\mathcal{W} = \int_V \underbrace{\bar{\bar{\mathrm{P}}}}_{\bar{\bar{\mathcal{T}}}\bar{\bar{\mathrm{F}}}^{-T}} : \bar{\bar{\nabla}}_0 \delta\vec{u} \,\mathrm{dV} - \left(\int_V \vec{f_0} \cdot \delta\vec{u} \,\mathrm{dV} + \int_S \vec{t_0} \cdot \delta\vec{u} \,\mathrm{dS} \right) = 0 \quad \forall\, \delta\vec{u} \tag{4.5}$$

4.2 Discretization

The discretization is a important step for the finite volume method. In this part, we use interpolation based on nodal values for the value of displacement inside an element. This is done by using shape functions (N) for each element node a (N_a).

$$\vec{u}(\vec{X}) = \sum_{a \in e} N_a \left(\vec{\xi} \right) \cdot \vec{U}_a \tag{4.6}$$

$$\bar{\bar{\nabla}}_0 \vec{u}(\vec{X}) = \sum_{a \in e} \vec{\nabla}_0 N_a \left(\vec{\xi} \right) \otimes \vec{U}_a \tag{4.7}$$

In those, the shape function is given as a function of the position vector $\vec{\xi}$. This is due to the fact that these functions are defined for a reference coordinate space. For a line element this would be for $\xi \in [-1, 1]$, for example. It is then possible to use the element's transformation (Equation 4.8) to relate the reference system ($\vec{\xi}$) to the element's global coordinates ($\vec{X}$).

$$\vec{X} = \psi(\vec{\xi}) \qquad \Longrightarrow \qquad \vec{\xi} = \psi^{-1}(\vec{X}) \tag{4.8}$$

In addition, the Galerkin method is used, which means that the virtual field is given by:

$$\delta\vec{u}(\vec{X}) = \sum_{a\,\in\,e} N_a\left(\vec{\xi}\right) \cdot \delta\vec{U}_a \tag{4.9}$$

$$\bar{\bar{\nabla}}_0 \delta\vec{u}(\vec{X}) = \sum_{a\,\in\,e} \vec{\nabla}_0 N_a\left(\vec{\xi}\right) \otimes \delta\vec{U}_a \tag{4.10}$$

4.3 Elementary description

The total virtual work of the system can be given as the sum of the virtual work for each element as stated below. This means that the final equation will be the result of the sum of the contribution of each element. For that reason, it is possible to work with the element-wise contribution and then add them up (this will be used in the assembly process - Section 4.5).

$$\delta\mathcal{W} = \sum_{e} \delta\mathcal{W}^{(e)} = 0 \quad \forall\, \delta\vec{u}^{(e)} \tag{4.11}$$

Therefore, by analyzing the element contribution fro the virtual work, the following relation is obtained

$$\delta\mathcal{W}^{(e)} = \int_{V^{(e)}} \bar{\bar{\mathrm{P}}}^{(e)} : \bar{\bar{\nabla}}_0 \delta\vec{u}^{(e)}\,\mathrm{dV} - \left(\int_{V^{(e)}} \vec{f_0}^{\,(e)} \cdot \delta\vec{u}^{(e)}\,\mathrm{dV} + \int_{S^{(e)}} \vec{t_0}^{\,(e)} \cdot \delta\vec{u}^{(e)}\,\mathrm{dS}\right)$$

By using the discretized functions in the equation above, it then becomes:

$$\delta\mathcal{W}^{(e)} = \sum_{a\,\in\,e} \underbrace{\left[\underbrace{\int_{V^{(e)}} \bar{\bar{\mathrm{P}}}^{(e)} \cdot \vec{\nabla}_0 N_a\,\mathrm{dV}}_{\vec{R}^{(e)}_{int,a}} - \underbrace{\left(\int_{V^{(e)}} \vec{f_0}^{\,(e)} \cdot N_a\,\mathrm{dV} + \int_{S^{(e)}} \vec{t_0}^{\,(e)} \cdot N_a\,\mathrm{dS}\right)}_{\vec{R}^{(e)}_{ext,a}}\right]}_{\vec{R}^{(e)}_a} \cdot \delta\vec{U}_a \tag{4.12}$$

In this equation the term $\vec{R}^{(e)}_a$ is the residual of the element e relative to node a. The goal of the finite element method is to find the values of $\vec{U}_a$ such that this residual is zero for all nodes a.

4.4 Numerical Integration

To solve for the integrals in the equations above, a numerical integration strategy is used. This is done with the Gauss quadrature framework.

Before using this process, one additional development needs to be done to Equation 4.12. As mentioned before, the shape functions are written in the space of the reference element. Therefore, it is useful to also integrate in this coordinate system. To make this change in integration variable, a volume and surface Jacobian for the $V \longrightarrow \hat{V}$ and $S \longrightarrow \hat{S}$ transformations are used and are represented by J_V and J_S, respectively.

$$
\begin{aligned}
\vec{R}^{(e)}_{int,a} &= \int_{V^{(e)}} \bar{\bar{\mathrm{P}}}^{(e)} \cdot \vec{\nabla}_0 N_a \,\mathrm{dV} = \\
&= \int_{\hat{V}^{(e)}} \bar{\bar{\mathrm{P}}}^{(e)} \left(\psi^{(e)}\left(\vec{\xi}\right)\right) \cdot \vec{\nabla}_0 N_a \left(\vec{\xi}\right) \cdot \mathrm{J}^{(e)}_V \mathrm{d\hat{V}} \\
\vec{R}^{(e)}_{ext,a} &= \int_{V^{(e)}} \vec{f}_0^{\,(e)} \cdot N_a \,\mathrm{dV} + \int_{S^{(e)}} \vec{t}_0^{\,(e)} \cdot N_a \,\mathrm{dS} = \\
&= \int_{\hat{V}^{(e)}} \vec{f}_0^{\,(e)} \left(\psi^{(e)}\left(\vec{\xi}\right)\right) \cdot N_a \left(\vec{\xi}\right) \cdot \mathrm{J}^{(e)}_V \mathrm{d\hat{V}} + \int_{\hat{S}^{(e)}} \vec{t}_0^{\,(e)} \left(\psi^{(e)}\left(\vec{\xi}\right)\right) \cdot N_a \left(\vec{\xi}\right) \cdot \mathrm{J}^{(e)}_S \mathrm{d\hat{S}}
\end{aligned}
$$

Now, by using the Gauss quadrature's approximation[4.1], it is possible to write:

$$
\vec{R}^{(e)}_{int,a} \approx \sum_g \omega_g \cdot \bar{\bar{\mathrm{P}}}^{(e)} \left(\psi^{(e)}\left(\vec{\xi_g}\right)\right) \cdot \vec{\nabla}_0 N_a \left(\vec{\xi_g}\right) \cdot \mathrm{J}^{(e)}_V \tag{4.13}
$$

$$
\vec{R}^{(e)}_{ext,a} \approx \sum_g \omega_g \cdot \vec{f_0^{(e)}} \left(\psi^{(e)}\left(\vec{\xi_g}\right)\right) \cdot N_a \left(\vec{\xi_g}\right) \cdot \mathrm{J}^{(e)}_V + \sum_{g_s} \vec{t_0^{(e)}} \left(\psi^{(e)}\left(\vec{\xi_{g_s}}\right)\right) \cdot N_a \left(\vec{\xi_{g_s}}\right) \cdot \mathrm{J}^{(e)}_S \tag{4.14}
$$

4.5 Assembly

The final step is to assemble the contribution of each element to obtain the global system of equations. This is done by summing the contribution of each element connected to node a. This results in a system with n_{nodes} vector equations with a varying from 1 to n_{nodes}. So, the total number or equations is $n_{nodes} \cdot dim$, where dim is the number of dimensions of the problem. The final system of equations is, then, given by:

$$
\sum_{e \in E_a} \left[\vec{R}^{(e)}_{int,a} - \vec{R}^{(e)}_{ext,a}\right] = \vec{0} \quad \forall\, a \in \{1, 2, ..., n_{nodes}\} \tag{4.15}
$$

Where E_a is the set of element that are connected to node a. This results in a system of equations that can be solved for the unknowns $\vec{U}_a$.

4.6 Boundary Conditions

In this finite element solver, three types of external effects can be applied: volumetric forces, Dirichlet boundary conditions and Neumann boundary conditions. The first are forces that act

[4.1] Note that the sum notation is used explicitly in the quadrature, while index notation is expressed implicitly for the remaining indices

on the volume of the body, such as gravity or body forces. They are represented by volumetric integral in external virtual work term (Equation 4.14). This is not a boundary condition, but simply an external effect that needs to be considered in the formulation. Therefore, only Dirichlet and Neumann boundary conditions are considered in this section.

4.6.1 Dirichlet Boundary Conditions

These are conditions that specify the value of the displacement at certain nodes. They are applied by directly setting the value of $\vec{U}_a$ for the nodes where the condition is applied. For the non linear solver this will be automaticaly taken into account in the residual, while for the linear solver it will be done by modifying the stiffness matrix and the force vector (disscussed below in Section 5.1).

4.6.2 Neumann Boundary Conditions

These are conditions that specify the value of the traction at certain nodes. They are applied by modifying surface integral of the external virtual work term (Equation 4.14) to include the contribution of the specified tractions at the nodes where the condition is applied. This is done by adding the contribution of the specified tractions to the force vector in the global system of equations.

Many types of Neumann boundary conditions can be applied, such as point loads, distributed loads, pressure, etc. Each of them will have a different contribution to the force vector. Below, the contribution of some of these types of Neumann boundary conditions is described:

— **To do**

Distributed load

Pressure

Chapter 5

Linear Solution

This chapter is dedicated to the solution of the linear system of equations obtained from the finite element formulation. This approach is used to the linear materials presented in Section 3.1. The final system of equations has the form given by Equation 5.1.

$$[K] \cdot \{U\} = \{F\} \tag{5.1}$$

In this case, $\bar{\bar{K}}$ is the stiffness matrix, $\vec{U}$ nodal displacement vector - unknown - and $\vec{F}$ is the vector of external forces. In order to obtain this system, the Equation 4.15 will serve as a starting point. The residual of the internal virtual work ($\vec{R}^{(e)}_{int,a}$) will be used to generate the $\bar{\bar{K}} \cdot \vec{U}$ term, while the residual of the external virtual work ($\vec{R}^{(e)}_{ext,a}$) will be used to generate the $\vec{F}$ term.

To obtain the stiffness matrix, we need to separate the nodal displacements from the rest of the terms in order to obtain the $\bar{\bar{K}}$ and $\vec{U}$ vector. In order to do this, the linearity of the material along with the hypothesis of small displacements (Equation 3.1) are used. We will, first, developthe stress - strain relation:

$$\bar{\bar{\sigma}} = \bar{\bar{\bar{\bar{\mathcal{C}}}}} : \bar{\bar{\varepsilon}} \Longrightarrow \bar{\bar{\sigma}}_{ij} = \mathcal{C}_{ijkl} \cdot \varepsilon_{kl}$$

Additionally, from the definition of strain (Equation 3.2) and the finite element discretization (Equation 4.7), the following relation is obtained:

$$\begin{aligned} \varepsilon_{kl} &= \frac{1}{2}\left(\nabla_0 u_{kl} + \nabla_0 u_{lk}\right) = \frac{1}{2}\left(\vec{\nabla}_0 N_b\left(\vec{\xi}\right) \otimes \vec{U}_b + \vec{U}_b \otimes \vec{\nabla}_0 N_b\left(\vec{\xi}\right)\right) \Longrightarrow \\ \varepsilon_{kl} &= \frac{1}{2}\left(\nabla_0 N_{bk}\left(\vec{\xi}\right) \cdot \delta_{lm} + \nabla_0 N_{bl}\left(\vec{\xi}\right) \cdot \delta_{km}\right) \cdot U_{bm} = B_{klbm}\left(\vec{\xi}\right) \cdot U_{bm} \end{aligned} \tag{5.2}$$

So, it is possible to use Equation 4.13[5.1] to obtaint the stiffness matrix, which is stated in Equation 5.3.

[5.1] Note that in the original reference, $\vec{R}^{(e)}_{int,a}$ is written as a vector, while in this section its dimensions are unpacked with the additional indice i. The same is done for the gradient of the shape functions.

$$R_{int,ai}^{(e)} \approx \left[\sum_g \omega_g \cdot \mathcal{C}_{ijkl} \cdot B_{klbm}\left(\vec{\xi}_g\right) \cdot \nabla_0 N_{aj}\left(\vec{\xi}_g\right) \cdot \mathrm{J}_V^{(e)} \right] \cdot U_{bm} \Longrightarrow$$

$$R_{int,ai}^{(e)} \approx K_{aibm}^{(e)} \cdot U_{bm} \qquad \text{where} \qquad K_{aibm}^{(e)} = \sum_g \omega_g \cdot \mathcal{C}_{ijkl} \cdot B_{klbm}\left(\vec{\xi}_g\right) \cdot \nabla_0 N_{aj}\left(\vec{\xi}_g\right) \cdot \mathrm{J}_V^{(e)} \tag{5.3}$$

Additionally, the external force vector is obtained by summing the contribution of each element to the external virtual work, which is done by the expression in Equation 4.14. Once the assembly process is done, the following system will be solved [5.2]:

$$K_{aibm} \cdot U_{bm} = F_{ai} \tag{5.4}$$

5.1 Dirichlet Boundary Conditions

These conditions are applied by modifying the stiffness matrix and the external force vector. For that each degree of freedom is divided the each degree of freedom into two sets: the set of free degrees of freedom (F) and the set of fixed degrees of freedom (D). The system of equations can then be written as stated in Equation 5.5.

$$\begin{bmatrix} K_{FF} & K_{FD} \\ K_{DF} & K_{DD} \end{bmatrix} \cdot \begin{bmatrix} U_F \\ U_D \end{bmatrix} = \begin{bmatrix} F_F \\ F_D \end{bmatrix} \tag{5.5}$$

From this point, the solver will initially solve for the free degrees of freedom (U_F in Equation 5.6), and then, the reactions at the fixed degrees of freedom can be obtained from F_D with Equation 5.7. This means that it possible to isolate the reactions by subtracting the virtual external work terms found above from F_D.

$$K_{FF} \cdot U_F = F_F - K_{FD} \cdot U_D \tag{5.6}$$

$$F_D = K_{DF} \cdot U_F + K_{DD} \cdot U_D \tag{5.7}$$

[5.2] It is important to note that the indices a, m and b, i represents the number of nodes and dimensions of the problem, respectively. To solve it numerically, the solver will need to combine them into a single index: $n_{nodes} \cdot dim$.

Chapter 6

Non Linear Solution

6.1 Newton-Raphson Method

The Newton-Raphson method is an iterative method used to solve non linear equations. For the case of a 1D equation, the formulation is given by:

$$f(x) = 0$$

The ideia is to start with an initial guess for the solution and then iteratively improve it until convergence is achieved. The method is based on the Taylor series expansion of the function around the current guess. The next guess is then given by:

$$x^{k+1} = x^k + \Delta x^k \qquad \text{where} \qquad \Delta x^k = -\frac{f(x^k)}{f'(x^k)} \tag{6.1}$$

As we expand this formulation, it is possible to solve for a general f that can be a scalar, vector, tensor or any type of function with a vector $\vec{x}$ as input:

$$f\left(\vec{x}\right) = 0$$

For this case, the definition of a directional derivative is needed. This can be defined as:

$$Df\left(\vec{x}\right)\left[\vec{u}\right] = \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} f\left(\vec{x} + \epsilon\vec{u}\right) \tag{6.2}$$

Therefore, the next guess for the solution is given by:

$$x^{k+1} = x^k + \Delta x^k \qquad \text{where} \qquad Df\left(\vec{x}^k\right)\left[\Delta\vec{x}^k\right] = -f\left(\vec{x}^k\right) \tag{6.3}$$

6.2 Application to Finite Elements

In order to apply the Newton-Raphson method to the finite element method, the equation to be solved is stated in 4.5. This means that the function to be solved is the virtual work of the system, and hence, the firts step is to compute the directional derivative of the virtual work with respect to a displacement increment $\Delta\vec{u}$:

$$D\delta\mathcal{W}\left(\vec{\phi},\, \delta\vec{u}\right)\,[\Delta\vec{u}] =$$

$$= \underbrace{\int_V D\bar{\bar{\mathrm{P}}}\,[\Delta\vec{u}] : \bar{\bar{\nabla}}_0\delta\vec{u}\,\mathrm{dV}}_{D\delta\mathcal{W}^{int}\,[\Delta\vec{u}]} - \underbrace{\left(\int_V D\vec{f_0}\,[\Delta\vec{u}]\cdot\delta\vec{u}\,\mathrm{dV} + \int_S D\vec{t_0}\,[\Delta\vec{u}]\cdot\delta\vec{u}\,\mathrm{dS}\right)}_{D\delta\mathcal{W}^{ext}\,[\Delta\vec{u}]} = 0 \quad \forall\,\delta\vec{u} \tag{6.4}$$

To solve for the next step, the virtual work of the system is composed of the sum of the internal virtual work and the external virtual work. Then, the directional derivative of each of them is computed separately. However, the directional derivative of the virtual work has the following form:

$$D\delta\mathcal{W}\,[\Delta\vec{u}] = \delta U_{aj}\cdot K_{ajbl}\cdot\Delta U_{bl}$$

The goal with this section is to find the tangent tensor K_{ajbl} for each combination of nodes a and b for both internal and external virtual work contributions. Once this is done, the next step is to solve for the increment of displacement $\Delta\vec{U}$ by solving the following system of equations:

$$D\delta\mathcal{W}\,[\Delta\vec{u}] = -\delta\mathcal{W} \qquad \Longrightarrow \qquad \delta U_{aj}\cdot K_{ajbl}\cdot\Delta U_{bl} = [-R_{aj}]\cdot\delta U_{aj} \qquad \forall\,\delta U_{aj} \Longrightarrow$$
$$\underbrace{[K_{ajbl}\cdot\Delta U_{bl} + R_{aj}]}_{0}\cdot\delta U_{aj} = 0 \qquad \forall\,\delta U_{aj} \tag{6.5}$$

6.2.1 Internal virtual work

The goal of this section is to compute the directional derivative of the internal virtual work. For that, the directional derivative of $\bar{\bar{P}}$ is initially evaluated. This is done by using the chain rule of differentiation:

$$D\bar{\bar{\mathrm{P}}}\,[\Delta\vec{u}] = \frac{\partial\bar{\bar{\mathrm{P}}}}{\partial\bar{\bar{F}}} : D\bar{\bar{\mathrm{F}}}\,[\Delta\vec{u}] = \frac{\partial\bar{\bar{\mathrm{P}}}}{\partial\bar{\bar{\mathrm{F}}}} : \bar{\bar{\nabla}}_0\Delta\vec{u} \tag{6.6}$$

Then, using that into Equation 6.4, the following expression is obtained for the directional derivative of the internal virtual work:

$$D\delta\mathcal{W}^{int}\,[\Delta\vec{u}] = \int_V D\bar{\bar{\mathrm{P}}}\,[\Delta\vec{u}] : \bar{\bar{\nabla}}_0\delta\vec{u}\,\mathrm{dV} = \int_V \bar{\bar{\nabla}}_0\delta\vec{u} : \frac{\partial\bar{\bar{\mathrm{P}}}}{\partial\bar{\bar{\mathrm{F}}}} : \bar{\bar{\nabla}}_0\Delta\vec{u}\,\mathrm{dV} \tag{6.7}$$

With this expression it is possible to isolate the tangent tensor $\overset{\equiv}{\overset{\equiv}{K}}$. This is done by using the definition of the finite element discretization of the displacement field. The final expression for the tangent tensor $\overset{\equiv}{\overset{\equiv}{K}}$ is given by:

$$D\delta\mathcal{W}^{int}\left[\Delta\vec{u}\right] = \int_V \overset{\equiv}{\nabla}_0 \delta\vec{u} : \frac{\partial \overset{\equiv}{\mathrm{P}}}{\partial \overset{\equiv}{\mathrm{F}}} : \overset{\equiv}{\nabla}_0 \Delta\vec{u}\,\mathrm{dV} = \int_V \left(\vec{\nabla}_0 N_a \otimes \delta\vec{U}_a\right) : \frac{\partial \overset{\equiv}{\mathrm{P}}}{\partial \overset{\equiv}{\mathrm{F}}} : \left(\vec{\nabla}_0 N_b \otimes \Delta\vec{U}_b\right)\,\mathrm{dV} \Longrightarrow$$

$$D\delta\mathcal{W}^{int}\left[\Delta\vec{u}\right] = \delta U_{aj} \cdot \underbrace{\left[\int_V \nabla_0 N_{ai} \cdot \frac{\partial \mathrm{P}_{ij}}{\partial \mathrm{F}_{kl}} \cdot \nabla_0 N_{bk}\,\mathrm{dV}\right]}_{K_{ajbl}} \cdot \Delta U_{bl} = \delta U_{aj} \cdot K_{ajbl} \cdot \Delta U_{bl}$$

The derivative of $\overset{\equiv}{\mathrm{P}}$ with respect to $\overset{\equiv}{\mathrm{F}}$, called the mixed elasticity tangent tensor, is expressed as a function of the material model used. The chain rule of differentiation is then used to express it as a function of the material tangent tensor $\overset{\equiv}{\overset{\equiv}{\mathcal{C}}}$:

$$\frac{\partial \overset{\equiv}{\mathrm{P}}}{\partial \overset{\equiv}{\mathrm{F}}} = \frac{\partial\left(\overset{\equiv}{\mathrm{F}} \cdot \overset{\equiv}{\mathrm{S}}\right)}{\partial \overset{\equiv}{\mathrm{F}}} = \frac{\partial \overset{\equiv}{\mathrm{F}}}{\partial \overset{\equiv}{\mathrm{F}}} \cdot \overset{\equiv}{\mathrm{S}} + \overset{\equiv}{\mathrm{F}} \cdot \frac{\partial \overset{\equiv}{\mathrm{S}}}{\partial \overset{\equiv}{\mathrm{F}}} = \frac{\partial \overset{\equiv}{\mathrm{F}}}{\partial \overset{\equiv}{\mathrm{F}}} \cdot \overset{\equiv}{\mathrm{S}} + \overset{\equiv}{\mathrm{F}} \cdot \underbrace{\frac{\partial \overset{\equiv}{\mathrm{S}}}{\partial \overset{\equiv}{\mathrm{E}}}}_{\overset{\equiv}{\overset{\equiv}{\mathcal{C}}}} \cdot \frac{\partial \overset{\equiv}{\mathrm{E}}}{\partial \overset{\equiv}{\mathrm{F}}} \tag{6.8}$$

Two of the unknown terms ($\frac{\partial \overset{\equiv}{\mathrm{F}}}{\partial \overset{\equiv}{\mathrm{F}}}$ and $\frac{\partial \overset{\equiv}{\mathrm{E}}}{\partial \overset{\equiv}{\mathrm{F}}}$) are purely kinematic and can be evaluated generally for any material model:

$$\frac{\partial \overset{\equiv}{\mathrm{F}}}{\partial \overset{\equiv}{\mathrm{F}}} \rightarrow \frac{\partial \mathrm{F}_{ij}}{\partial \mathrm{F}_{kl}} = \delta_{ik} \cdot \delta_{jl} \tag{6.9}$$

$$\frac{\partial \overset{\equiv}{\mathrm{E}}}{\partial \overset{\equiv}{\mathrm{F}}} = \frac{\partial\left(\frac{1}{2}\left(\overset{\equiv}{\mathrm{F}}^T \cdot \overset{\equiv}{\mathrm{F}} - \overset{\equiv}{I}\right)\right)}{\partial \overset{\equiv}{\mathrm{F}}} = \frac{1}{2}\left(\frac{\partial \overset{\equiv}{\mathrm{F}}^T}{\partial \overset{\equiv}{\mathrm{F}}} \cdot \overset{\equiv}{\mathrm{F}} + \overset{\equiv}{\mathrm{F}}^T \cdot \frac{\partial \overset{\equiv}{\mathrm{F}}}{\partial \overset{\equiv}{\mathrm{F}}}\right) \rightarrow \frac{\partial \mathrm{E}_{ij}}{\partial \mathrm{F}_{kl}} = \frac{1}{2}\left(\delta_{il} \cdot \mathrm{F}_{kj} + \delta_{jl} \cdot \mathrm{F}_{ki}\right) \tag{6.10}$$

However, the tensor $\overset{\equiv}{\overset{\equiv}{\mathcal{C}}}$ is the material tangent tensor, which is a function of the material model used. This is the information used in the code that each non linear material needs to provide. For that reason, the expression of $\overset{\equiv}{\overset{\equiv}{\mathcal{C}}}$ for each non linear material model is described below:

Saint Venant - Kirchhoff:

$$\overset{\equiv}{\overset{\equiv}{\mathcal{C}}} = \frac{\partial \overset{\equiv}{\mathrm{S}}}{\partial \overset{\equiv}{\mathrm{E}}} = \frac{\partial\left(\lambda\, tr(\overset{\equiv}{\mathrm{E}})\overset{\equiv}{\mathrm{I}} + 2\mu\,\overset{\equiv}{\mathrm{E}}\right)}{\partial \overset{\equiv}{\mathrm{E}}} \Longrightarrow$$

$$\mathcal{C}_{ijkl} = \lambda\,\delta_{ij}\delta_{kl} + \mu\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right) \tag{6.11}$$

Compressible Neo-Hookean:

$$\bar{\bar{\bar{\bar{\mathcal{C}}}}} = \frac{\partial \bar{\bar{\mathrm{S}}}}{\partial \bar{\bar{\mathrm{E}}}} = \frac{\partial \left(\mu \left(\bar{\bar{\mathrm{I}}} - \bar{\bar{\mathrm{C}}}^{-1} \right) - \lambda \ln \mathrm{J} \, \bar{\bar{\mathrm{C}}}^{-1} \right)}{\partial \bar{\bar{\mathrm{E}}}} \Longrightarrow$$

$$\mathcal{C}_{ijkl} = (\mu - \lambda \ln \mathrm{J}) \cdot \left(\mathrm{C}^{-1}_{ik} \cdot \mathrm{C}^{-1}_{jl} + \mathrm{C}^{-1}_{il} \cdot \mathrm{C}^{-1}_{jk} \right) + \lambda \cdot \mathrm{C}^{-1}_{ij} \cdot \mathrm{C}^{-1}_{kl} \tag{6.12}$$

6.2.2 External virtual work

To do

Chapter 7

The code

7.1 Geometry

7.2 Mesh

7.3 Material

7.4 Element

7.5 Model

7.6 Boundary Conditions

7.7 Post Processing